Paleomagnetism, U–Pb geochronology, and geochemistry of Marathon dykes, Superior Province, and comparison with the Fort Frances swarm[1]

Kenneth L. Buchan, Henry C. Halls, and James K. Mortensen

Abstract: We report the first detailed study of the paleomagnetism, U–Pb geochronology and major element geochemistry of Paleoproterozoic north-trending Marathon dykes north of Lake Superior. The paleomagnetic and geochemical results demonstrate that Marathon dykes can be divided into two subsets, one of normal magnetic polarity, the other of reverse polarity. Normal and reverse Marathon paleomagnetic poles, at 43°N, 196°E (dm = 9°, dp = 7°, number of dykes N = 16) and 51°N, 175°E (dm = 9°, dp = 6°, N = 12), respectively, are statistically distinct and may indicate different ages of normal and reverse dyke emplacement. A U–Pb baddeleyite age of 2121^{+14}_{-7} Ma has been obtained at a normally magnetized Marathon paleomagnetic site. The reversely magnetized Marathon dykes are undated, but have a paleopole rather close to that of the reversely magnetized 2076^{+5}_{-4} Ma Fort Frances dykes and major element geochemical signatures as portrayed on Jensen plots that are identical to those of the Fort Frances swarm. Therefore, reverse Marathon and Fort Frances dykes could define a giant radiating dyke swarm focused south of Lake Superior, supporting models that associate these dykes with Paleoproterozoic rifting along the southern margin of the Superior Province. The Marathon and Fort Frances paleopoles continue a northwesterly trend in southern Superior Province paleopoles, which has recently been defined by results for 2216^{+8}_{-4} Ma Senneterre dykes and 2167 ± 2 Ma Biscotasing dykes. This trend contrasts with previous widely used polar wander paths for the same period that young in the opposite direction and illustrates the importance of collaborative studies of paleomagnetism and U–Pb geochronology.

Résumé : Nous présentons les résultats de la première étude détaillée de paléomagnétisme, géochronologie U–Pb et géochimie des éléments majeurs des dykes de Marathon, de direction nord et d'âge Paléoprotérozoïque, qui apparaissent au nord du lac Supérieur. Les données paléomagnétiques et géochimiques révèlent qu'on peut séparer les dykes de Marathon en deux sous-groupes : un avec une polarité magnétique normale, et l'autre avec polarité inverse. Les pôles paléomagnétiques de polarités normale et inverse des dykes de Marathon sont à 43°N, 196°E (dm = 9°, dp = 7°, nombre de dykes N = 16) et 51°N, 175°E (dm = 9°, dp = 6°, N = 12), respectivement; ils sont statistiquement distincts et il y a eu vraisemblablement la mise en place de ces dykes de polarités normale et inverse à des périodes différentes. Un âge U–Pb sur baddeleyite de 2121^{+14}_{-7} Ma a été obtenu pour un site paléomagnétique d'aimantation normale des dykes de Marathon. Les dykes d'aimantation inverse ne sont pas datés, mais leurs paléopôles sont localisés très près de ceux fournis par les dykes de Fort Frances d'aimantation inverse datés de 2076^{+5}_{-4} Ma, et leurs signatures géochimiques pour les éléments majeurs dans les diagrammes de Jensen sont identiques à celles de l'essaim de dykes de Fort Frances. Pour ces raisons, il est possible d'intégrer les dykes d'aimantation inverse de Marathon et de Fort Frances au gigantesque essaim de dykes rayonnant et centré au sud du lac Supérieur, ce qui plaide en faveur des modèles qui associent ces dykes au développement d'un rift paléoprotérozoïque le long de la marge méridionale de la Province du Supérieur. Les paléopôles des dykes de Marathon et de Fort Frances sont en continuité avec la direction nord-ouest de paléopôles observés dans la région méridionale de la Province du Supérieur, récemment définie par les mesures sur les dykes de Senneterre datés de 2216^{+8}_{-4} Ma et les dykes de Biscotasing âgés de 2167 ± 2 Ma. Cette direction est en désaccord avec les courbes de la dérive des pôles couramment utilisées antérieurement pour la même période qui présent sa face plus jeune en direction opposée, ce qui démontre toute l'importance des études communes de paléomagnétisme et de géochronologie U–Pb.
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Introduction

Several diabase dyke swarms of Paleoproterozoic age form a fan-shaped distribution about Lake Superior (Fig. 1). Senneterre and Biscotasing dykes are located east of the lake, Marathon dykes to the north, and Fort Frances (or Kenora–Kabetogama) dykes to the west. It has been suggested that one or more of these swarms is associated with Paleoproterozoic rifting and the development of the Animikie Basin, which forms part of the Southern Province (Fig. 1) along the southern margin of the Superior Province (e.g., Southwick and Day 1983; Fahrig 1987).

Prior to the present study, the dykes east and west of Lake Superior were rather well characterized in terms of distribution, age, paleomagnetism, and geochemistry, whereas the Marathon dykes were little known. The Biscotasing swarm had a well-defined U–Pb age of 2167 ± 2 Ma (Buchan et al. 1993). The Senneterre swarm was dated at 2214 ± 12 Ma (Buchan et al. 1993), although further analysis has yielded a more precise age of 2216^{+8}_{-4} Ma (J.K. Mortensen, unpublished data). Both the Biscotasing and Senneterre swarms were known to carry primary paleomagnetic remanences (Buchan et al. 1993), and had been studied geochemically (Condie et al. 1987). The Fort Frances swarm was dated by the Rb–Sr technique at 2127 ± 67 Ma (Beck and Murthy 1982). More precise U–Pb dates of 2076^{+5}_{-4} and 2077^{+4}_{-3} Ma are reported in this study and in recent work by Wirth et al. (1995), respectively. A well-defined primary paleomagnetic pole (Halls 1986), detailed geochemistry (Southwick and Halls 1987; Schmitz et al. 1995), Nd isotopic studies (Wirth and Vervoort 1995), and tectonic setting (Southwick and Day 1983) had also been described for the Fort Frances swarm.

The age of the Marathon dykes (Fig. 2) was constrained only by K–Ar whole-rock dates ranging from 1950 to 1550 Ma and a K–Ar biotite date of 1705 Ma (Fahrig et al. 1986). Based on these data, Fahrig et al. (1986) suggested an age of intrusion >2000 Ma. In this paper, we report a U–Pb baddeleyite age of 2121^{+14}_{-7} Ma for a Marathon dyke.

Fahrig et al. (1965) reported paleomagnetic results from 12 samples collected at seven Marathon dyke sites. Each sample was demagnetized to peak alternating fields (AF's) of only 13 mT. More detailed measurements were later carried out at a single site near Geco mine at Manitouwadge (Fig. 2) (Fahrig et al. 1986). Evidence for the primary nature of the Marathon remanence was not obtained in these early studies.

The impetus for sampling Marathon dykes in the present study was twofold: (*i*) to characterize them in terms of distribution, paleomagnetism, U–Pb geochronology, and geochemistry, and (*ii*) to clarify their relationship to other neighbouring swarms and to rifting along the southern margin of the Superior Province. As a U–Pb age was not then available for the Fort Frances swarm, material was also collected from a Fort Frances dyke for dating and paleomagnetism. Following the recommendations of Buchan and Halls (1990), U–Pb geochronology was conducted at paleomagnetic sites, and a rigorous baked contact test was carried out to test for the primary nature of the remanence.

Geological setting

Paleoproterozoic Marathon diabase dykes trend approximately north over an area of at least 30 000 km^2 north of Lake Superior and east of Lake Nipigon (Fig. 2), where they intrude Archean rocks of the Superior Province. The extent of the swarm to the north and east of the area shown in Fig. 2 is unknown, because little work has been done to distinguish dykes of different swarms. To the southeast, Marathon dykes have not yet been identified beyond the sampling area, with the possible exception of two dykes located east of Lake Superior (sites 28 and 31 of Halls and Shaw 1988), which have paleomagnetic directions similar to those of Marathon dykes. To the west of the sampling area, Marathon dykes are overlain by Mesoproterozoic rocks of the Southern Province in the vicinity of Lake Nipigon.

Marathon dykes are well exposed in roadcuts and along lakeshores. Elsewhere, they are best traced from aeromagnetic data. They have subvertical dips, are usually less than about 50 m wide, and exhibit chilled margins with sharp contacts.

Dykes of several other swarms are thought to occur within the sampling area. Keweenawan diabase dykes of approximately 1100 Ma age trend east–west in the vicinity of Lake Superior. The 2167 Ma northeast-trending Biscotasing diabase dyke swarm has been studied in detail to the southeast and is thought to extend into the sampling area (H.C. Halls and K.L. Buchan, unpublished data). Finally, 2473–2446 Ma (Heaman 1995) Matachewan diabase dykes trend northwest across the region.

Sampling and laboratory procedure

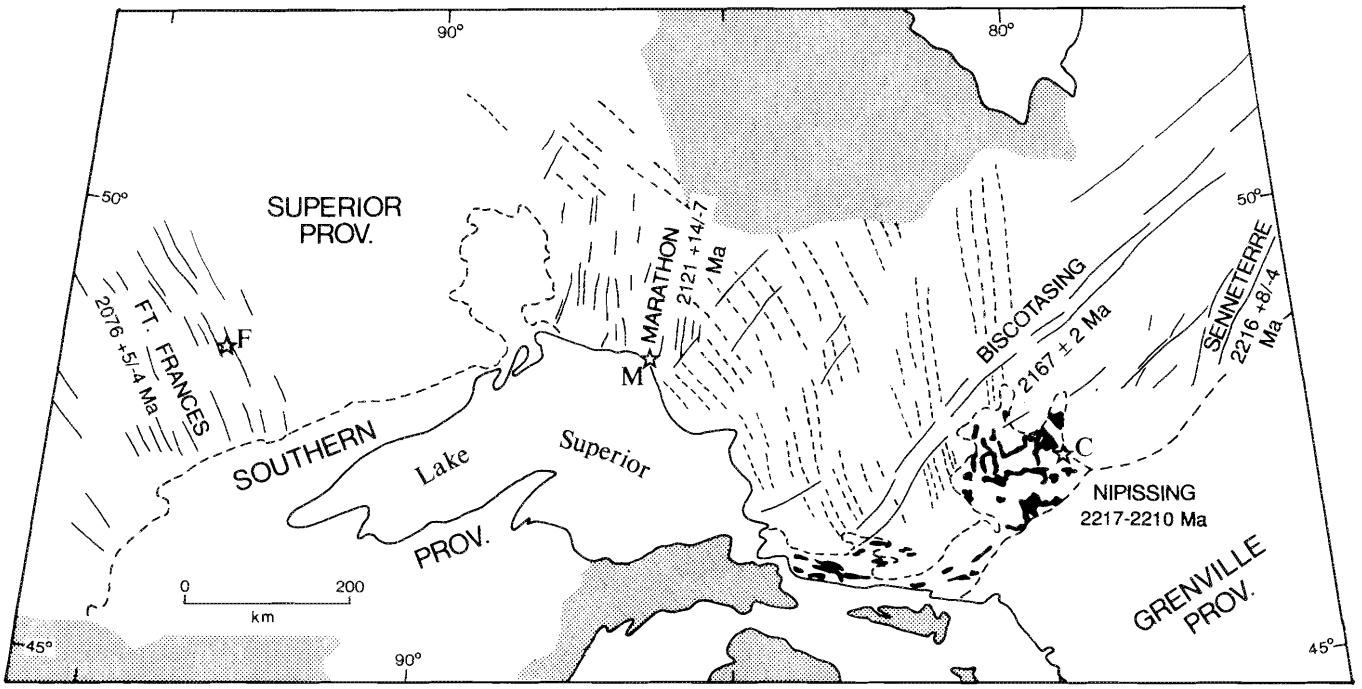
Five to twelve paleomagnetic block samples were oriented with sun and magnetic compasses at each of 32 sites (Fig. 2). Each sample was drilled in the laboratory to produce specimens with a diameter of 2.5 cm and a length of 2.2 cm. Specimens were measured using either a Schonstedt or modified DIGICO spinner magnetometer, and were subjected to stepwise AF demagnetization, thermal step heating, or a combination of AF demagnetization followed by thermal cleaning.

A baked contact test was carried out at site 29, where a north-trending Marathon dyke cuts a northwest-trending dyke of the Matachewan swarm. Samples were collected within the Marathon dyke, adjacent to the Marathon dyke in the host Matachewan dyke, and farther away in the Matachewan dyke.

A sample for U–Pb geochronology was collected from the coarse interior of a ~50 m wide Marathon dyke at paleomagnetic site 22, after a single baddeleyite fraction was separated from a small hand sample at this site from the earlier collection of Fahrig et al. (1965). Likewise, a sample for dating was obtained from a >30 m wide Fort Frances dyke located on the shore of Kakagi Lake at 49°14.5′N, 93°58.5′W, following the analysis of a small hand sample from the Fahrig et al. collection that indicated the presence of baddeleyite. As the Kakagi Lake site was not sampled in the earlier paleomagnetic study of Fort Frances dykes (Halls 1986), samples were collected for paleomagnetism. They yielded results similar to those of site 24 of Halls (1986).

Zircon and baddeleyite were separated from ~25 kg samples using conventional crushing, grinding, wet shaking table, heavy liquids, and magnetic separator methods. Individual fractions for analysis were selected on the basis of grain size and morphology. Analyses were carried out in the

Fig. 1. Paleoproterozoic dykes and sills of the Lake Superior region. Nipissing sills are coloured black. Only U–Pb ages are shown. The 2473–2446 Ma Matachewan dyke swarm is shown as broken lines. Stars labelled F, M, and C locate the towns of Fort Frances, Marathon, and Cobalt, respectively. Age references: Nipissing sills, Corfu and Andrews (1986) and Noble and Lightfoot (1992); Senneterre dykes, Buchan et al. (1993) and J.K. Mortensen (unpublished data); Biscotasing dykes, Buchan et al. (1993); Marathon and Fort Frances dykes (this study). Phanerozoic rocks are shaded.



Geochronology Laboratory at the Geological Survey of Canada (Ottawa), using methods described by Mortensen (1993) and Buchan et al. (1993). The small amount of baddeleyite recovered from each sample was too fine grained to permit the grains to be abraded prior to dissolution, or to be completely sure that the grains were free of secondary zircon (Heaman and LeCheminant 1993).

Major element geochemical data were obtained from chilled-margin samples of Marathon dykes following the same sampling procedure and measurement methodology as used previously for the Fort Frances dykes (Halls 1986; Southwick and Halls 1987). Both the Marathon and Fort Frances analyses were carried out by X-Ray Assay Laboratories of Don Mills, Ontario.

Paleomagnetic results

Site directions

After AF and (or) thermal demagnetization, 29 Marathon sites were found to be stably magnetized, and four were either unstable or lightning struck. In most cases, soft components, likely of viscous origin, were eliminated in AF's of 10–15 mT. Subsequent stepwise cleaning revealed a stable characteristic magnetization of normal or reverse polarity, directed down west-northwest or up southeast, respectively (Fig. 3). On AF demagnetization, the stable characteristic component is usually eliminated in AF's of 40–100 mT. At higher fields directions become scattered.

A combination of AF and thermal demagnetization proved effective in isolating the stable magnetization at some sites (Fig. 4). After removal of viscous components by the AF method, an improved definition of the characteristic component was obtained by subsequent thermal treatment. Blocking temperatures of the characteristic component fall in a range below 560 or 580°C, suggesting low-titanium titanomagnetite as the remanence carrier. In some sites the intersection of remagnetization circles (Bailey and Halls 1984), combined with stable end points where possible, defined the direction of the characteristic magnetization.

At sites 27 and 28, both normal and reversely magnetized components appear to coexist in some specimens (e.g., Fig. 5). Coercivities (or blocking temperatures) of the normal component may be either above or below those of the reverse one, so the relative age of the two magnetizations is uncertain.

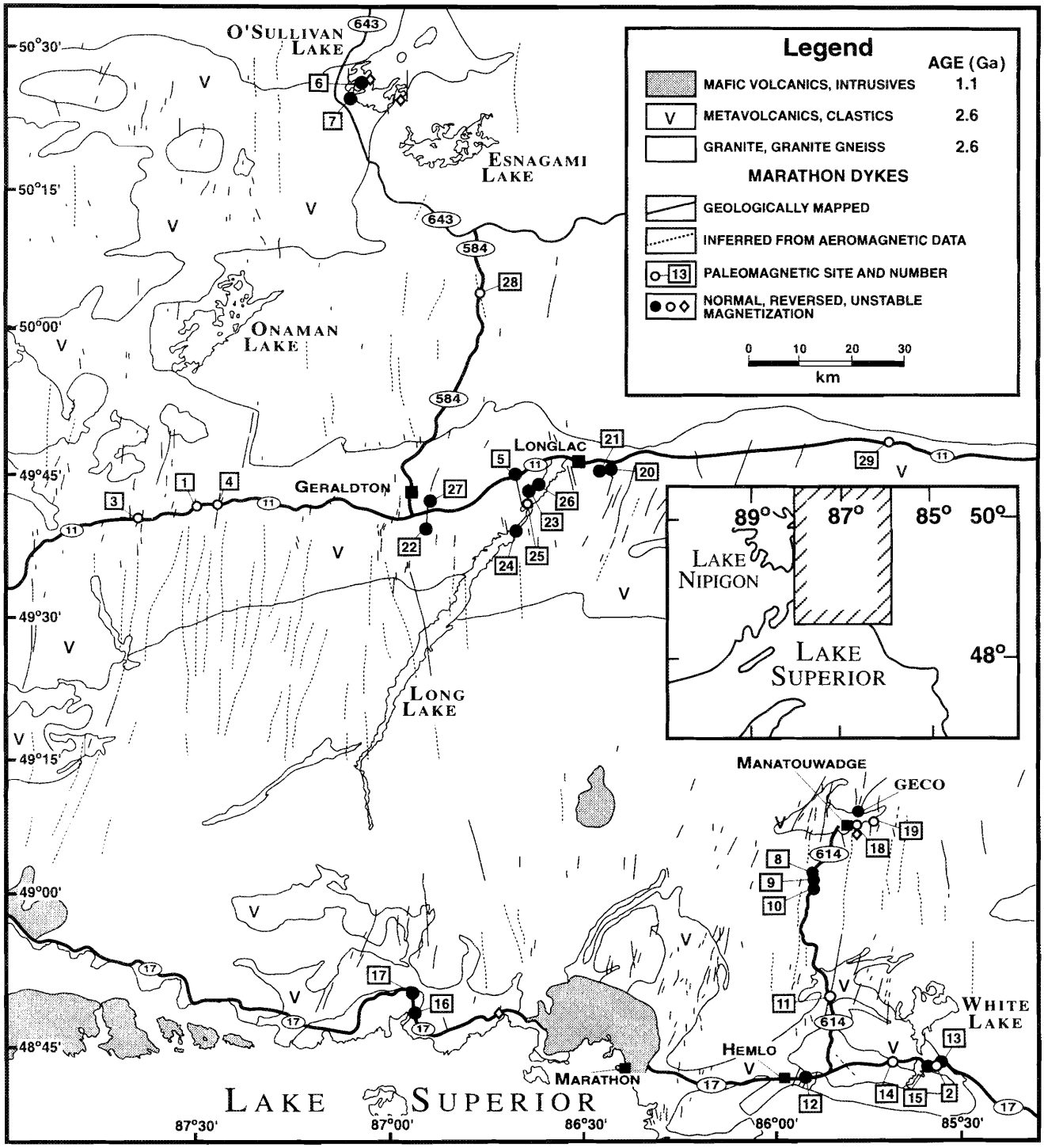
Marathon site mean directions are given in Table 1 and Fig. 6. The overall normal and reverse means are summarized in Table 2 along with corresponding pole positions. Of 29 stable sites, 12 are reversely magnetized and 17 normally magnetized. Sites 8–10 are located in close proximity along a single dyke, so their remanence directions have been combined in the calculation of the overall normal mean. In addition, a site near Geco mine (Fahrig et al. 1986) has been included in the statistics for the normal magnetization.

The mean direction of reverse Marathon dykes is only approximately antipodal to that of normal dykes (Fig. 6). A small but statistically significant asymmetry is observed at a confidence level of 95% (Watson 1956; McFadden and Lowes 1981). As discussed below, this may reflect an age difference between normally and reversely magnetized dykes.

Baked contact test

A positive baked contact test that satisfies the rigorous

Fig. 2. Map showing the Marathon dyke swarm, regional geological setting, and location of paleomagnetic sampling sites. A U–Pb age is reported from site 22 and a baked contact test form site 29. The shaded rocks are those associated with the 1.1 Ga Mid-Continent Rift (intrusive rocks on the mainland, predominantly volcanic rocks on the islands).



criteria of Buchan and Halls (1990) was obtained at site 29 (Figs. 7, 8). Here a 5 m wide reversely magnetized Marathon dyke with a trend of 10°E crosscuts a 13 m wide Matachewan dyke trending 325°E (Fig. 7*a*). The intersection of the dykes is well exposed in a prominent roadcut on Highway 11.

Samples 1–5 were collected in the younger Marathon dyke, samples 6–10 in the Matachewan dyke within a distance of 2.5 m of the Marathon dyke, and samples 11–14 in the Matachewan dyke at a distance of 6.8–9.6 m from the Marathon dyke (Fig. 7).

Both the Marathon samples and those collected in the Matachewan dyke close to the contact with the Marathon dyke carry a reverse Marathon remanence (Figs. 7*b*, 8*a*, 8*b*).

Fig. 3. Typical AF demagnetization results for Marathon dykes. (*a*, *b*) Normally magnetized specimens from sites 7 and 12, respectively, (*c*, *d*) Reversely magnetized specimens from sites 4 and 2, respectively. Closed and open circles on equal-area projections indicate downward and upward directions, respectively. Circles and triangles on vector diagrams represent projection onto the horizontal and vertical planes, respectively. Alternating fields are labelled in mT. Magnetization intensity units in A/m $\times$ 10^{-3}.

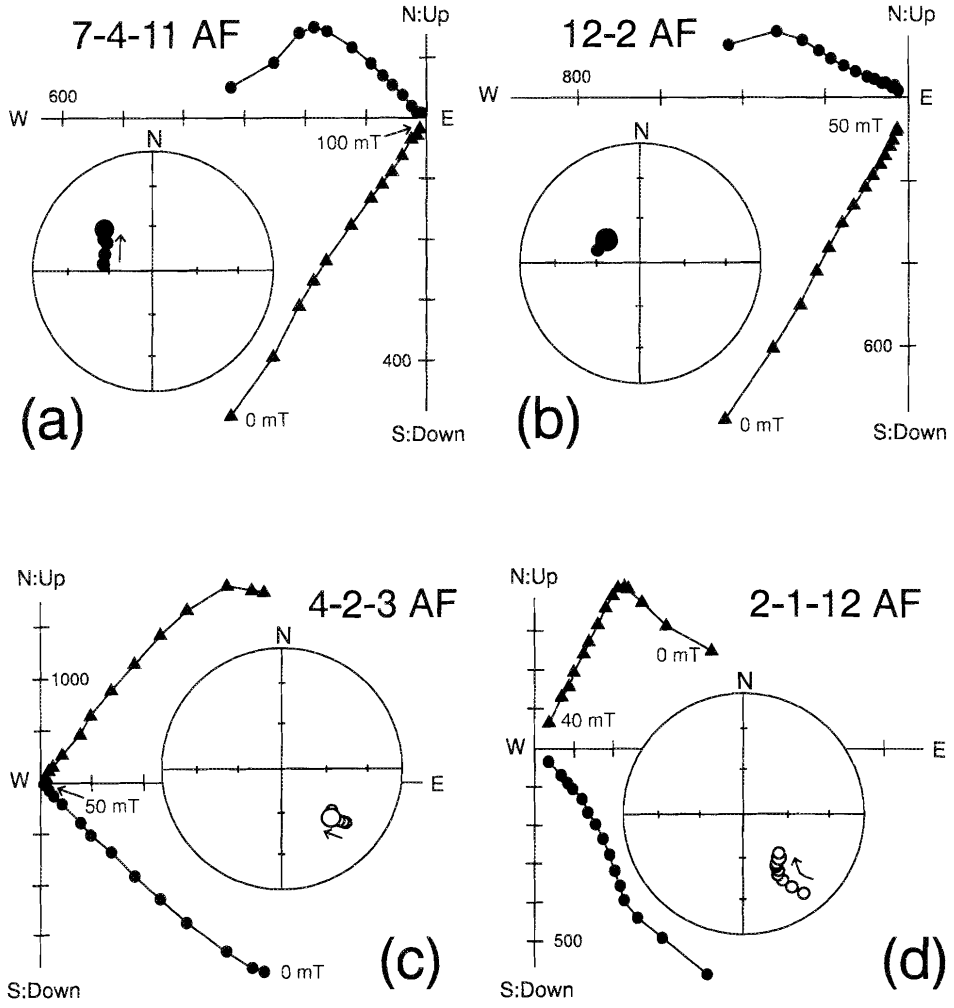


Fig. 4. Combined AF – thermal demagnetization results. (*a*) Reversely magnetized specimen from site 28. (*b*) Normally magnetized specimen from site 21. Symbols and explanations as in Fig. 3.

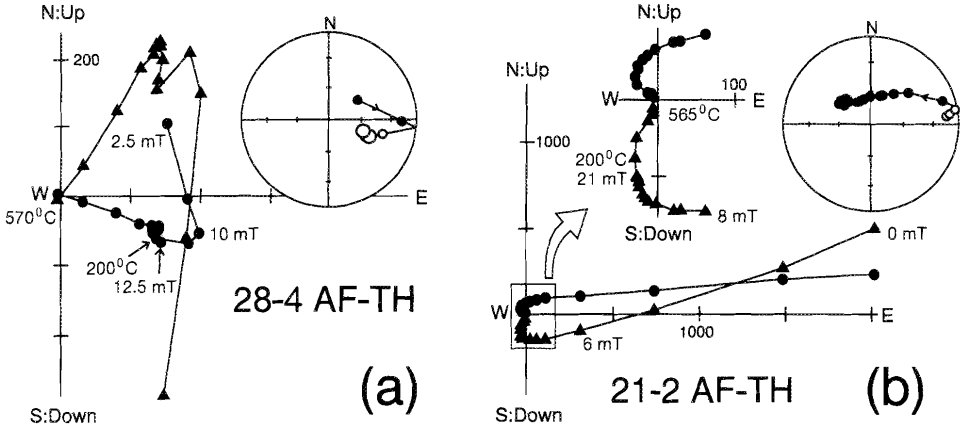
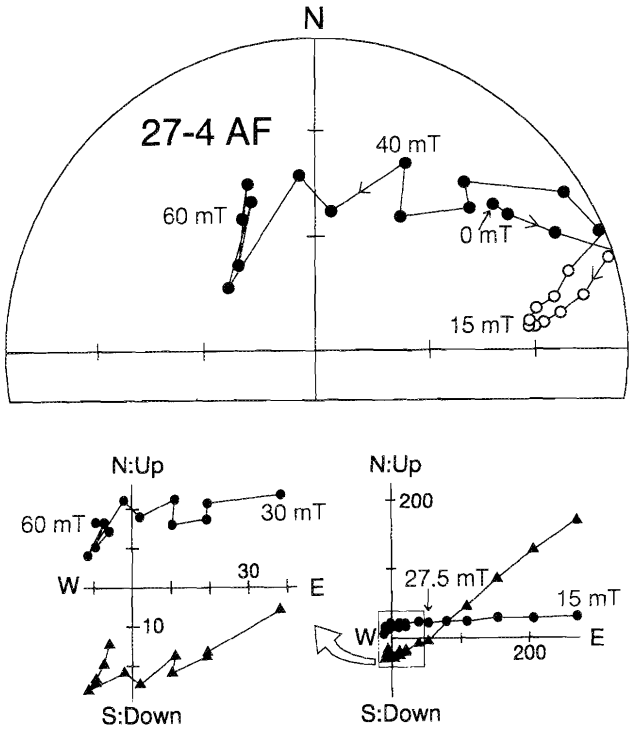


Fig. 5. Example of superimposed normal and reversed magnetization components from site 27 upon AF demagnetization. The vector diagram on the right shows a clearly defined reverse direction that is close to the Marathon direction, but the enlargement on the left shows that the normal component is poorly defined by comparison. Symbols and explanations as in Fig. 3.

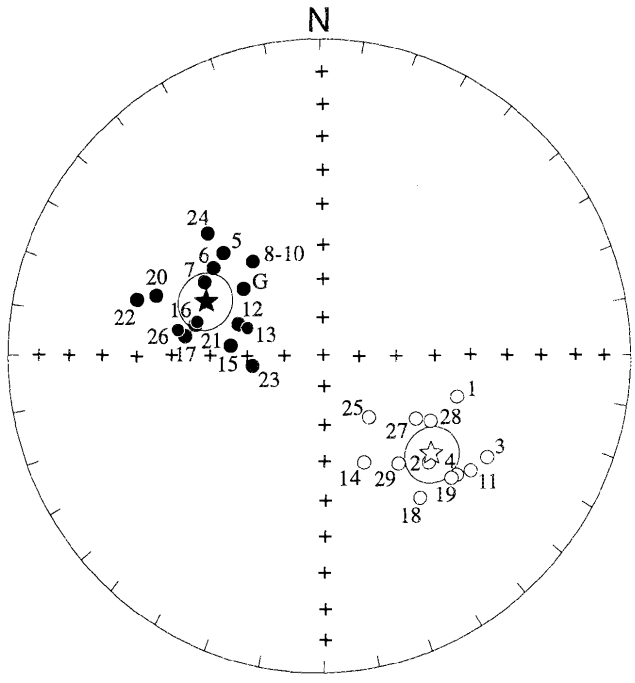


Matachewan dyke samples remote from the Marathon dyke carry a typical reverse Matachewan remanence, directed to the south-southwest with shallow negative inclination (Figs. 7*b*, 8*c*), indicating that the Matachewan samples close to the Marathon dyke were remagnetized at the time of Marathon dyke emplacement. This positive baked contact test demonstrates that the Marathon dyke remanence is primary, acquired at the time of intrusion.

U–Pb geochronology

Analytical data from the U–Pb dating study are given in Table 3, and results are shown graphically in Fig. 9. Baddeleyite separated from the normal Marathon dyke sample at site 22 consisted of fine reddish-brown cleavage flakes. A total of six fractions were analysed. Three fractions yielded relatively imprecise analyses. Five fractions gave U concentrations (136–396 ppm) typical of baddeleyite analyses reported by other workers. One fraction gave a much higher U concentration (1411 ppm); this may reflect either a naturally occurring variation in U concentration or, more likely, a weighing error due to the small sample size. The six analyses define a linear array, and range from 1.8 to 9.9% discordant (Fig. 9*a*). A Davis-type regression (Davis 1982) with expanded errors yields upper and lower intercept ages of 2121^{+14}_{-7} and 1054 Ma, respectively, with a probability of fit of 45.9%. The upper intercept is interpreted as the crystallization age of the sample, and the lower intercept reflects

Fig. 6. Site mean paleomagnetic directions (stars) for Marathon dykes with their ovals of 95% confidence. Circles are results for individual sites. G, Geco mine site of Fahrig et al. (1986). All data are listed in Table 1. Symbols and explanations as in Fig 3.



Pb loss of broadly Keweenawan or Grenvillian age.

Baddeleyite recovered from the Fort Frances dyke sample at Kakagi Lake was similar in appearance to that of the Marathon dyke sample. Zircon in this sample formed elongate prisms and rare skeletal grains with abundant inclusions, and broken fragments of these grains. The zircon was generally unzoned, and contained no visible cores. Two fractions of baddeleyite and four fractions of zircon were analysed. The analyses range from 5.3 to 11.7% discordant (Fig. 9*b*). A Davis-type regression (Davis 1982) is, therefore, not valid for this data set, and a modified York-II regression model (Parrish et al. 1987) was employed instead. The two baddeleyite fractions and three of the four zircon fractions define a linear array with calculated upper and lower intercept ages of 2076.3^{+5}_{-4} and 761 Ma, respectively, with a mean square of weighted deviates (MSWD) of 2.7. The fourth zircon fraction (G) falls slightly to the right of this regression, suggesting that it likely contained a minor component of older, inherited zircon, either as one or more xenocrysts or as "cryptic" cores that could not be detected visually. The upper intercept is taken as the best estimate for the crystallization age of the sample, and the discordance of the analyses is considered to reflect relatively young Pb loss. However, the possibility that the baddeleyite fractions contained a small proportion of secondary zircon as an alteration product cannot be ruled out. Our age of 2076^{+5}_{-4} for the Fort Frances dyke is identical to a baddeleyite–zircon age of 2077^{+4}_{-3} Ma recently reported from another dyke in the swarm (Wirth et al. 1995).

Although our Marathon dyke age is relatively imprecise, it is clear that normally magnetized Marathon dykes and

Table 1. Paleomagnetic results for sites in Marathon diabase dykes.

Site	Lat. (°N)	Long. (°W)	Trend (°E)	Dip	Width (m)	N	n_E	n_R	D (°)	I (°)	D' (°)	I' (°)	k	α_{95} (°)
1	49.69	87.49	45	70°E	23	6	4	1	107.6	−52.6	109.0	−51.4	220	5.6
2	48.72	85.60	356	85°W	9	8	3	4	135.7	−50.2	135.4	−50.6	210	5.5
3	49.67	87.66	354	~90°	24	11	4	0	122.4	−38.3	123.7	−36.5	29	17.5
4	49.70	87.46	0	—	76	5	5	0	131.8	−42.0	132.9	−40.4	940	2.5
5	49.74	86.68	5	85°W	15	8	4	2	316.4	52.8	317.0	51.9	85	8.5
6	50.44	87.07	0	—	>6	6	4	1	309.3	53.5	310.6	52.0	190	6.0
7	50.44	87.10	350	>55°E	>5	8	6	1	301.8	52.0	303.2	50.5	140	5.4
8, 9, 10[a]	49.04	85.90	10	—	12−30	22	12	3	323.7	59.3	323.7	59.3	77	5.0
11	48.83	85.85	20	85°W	46	15	2	5	128.4	−40.1	128.3	−40.4	79	10.0
12	48.70	85.91	20	>85°E	>24	9	5	4	290.9	66.8	290.5	66.9	130	5.3
13	48.72	85.59	10−15	80°E	5.5	9	2	5	290.2	69.5	289.7	69.7	220	5.8
14	48.73	85.71	350	85°E?	24	9	6	1	159.5	−60.0	159.3	−60.3	47	9.8
15	48.72	85.60	10−20	80−85°E	12	9	6	2	276.5	65.4	275.9	65.6	140	5.2
16	48.81	86.94	30	85°E	10.5	9	5	0	287.9	55.3	288.3	54.8	100	7.7
17	48.85	86.96	28	80°W	29	12	7	0	278.4	53.7	279.0	53.1	140	5.1
18	49.13	85.79	~0	—	>3	5	4	0	146.4	−43.6	146.4	−43.6	216	6.3
19	49.14	85.75	0−10	—	>25	7	3	1	135.0	−42.6	134.9	−42.6	140	8.9
20	49.77	86.43	350	—	8−10	5	2	3	289.5	44.7	290.2	44.0	180	7.8
21	49.77	86.43	~0	—	>15	5	5	0	284.9	57.0	285.7	56.6	43	11.8
22[b]	49.18	86.91	~0	—	~50	6	4	0	286.3	38.2	287.0	37.3	234	6.0
23	49.72	86.63	N−NNW	—	>3	5	4	0	250.4	71.1	252.1	71.0	30	17.1
24	49.66	86.67	0[c]	—	>4	5	5	0	316.8	46.3	317.3	45.4	169	5.9
25	49.71	86.64	NNW	—	>5	5	5	0	144.5	−69.9	145.1	−69.3	401	3.8
26	49.23	86.61	~0	—	>3	5	5	0	279.2	51.6	279.8	51.1	31	14.0
27	49.21	86.90	~5[d]	—	~50	5	0	5	125.7	−59.5	129.0	−56.0	140	10/29[e]
28	50.06	86.76	~0	—	15	5	4	0	123.0	−56.3	124.0	−55.4	154	7.4
29[f]	49.80	85.69	10	90°	5	5	4	0	145.7	−54.8	145.9	−54.3	59	12.0
29B[f]	49.80	85.69	—	—	—	5	4	0	133.0	−53.6	133.3	−53.2	49	13.2
29U[f]	49.80	85.69	325	90°	13	4	4	0	202.3	−31.8			55	12.5
GECO[g]	~49.0	~85.8				6	6	0	310	64	310	64	28	12.9

Notes: Lat. and Long., latitude and longitude of the sampling sites; N, number of samples measured; n_E, number of sample Kirschvink (1980) analyses used in statistical calculation; n_R, number of sample remagnetization circles used in statistical calculation; D and I, declination and inclination of the remanent magnetization; D' and I', declination and inclination corrected to a common reference site at 49°N, 86°W; k, precision parameter; α_{95}, radius of the circle of 95% confidence about the mean direction. Statistics are based on AF demagnetization, except at sites 18, 20, 21, 26, 27, 28, and 29, in which both AF and combined AF−thermal data were used.

[a]Paleomagnetic data have been combined for sites 8, 9, and 10, which are located on a single dyke a short distance from each other.

[b]This site was dated by the U−Pb technique (see text).

[c]Trend from Kresz et al. (990).

[d]Trend from Ontario Department of Mines (1951).

[e]Minimum and maximum values of α_{95} calculated using Bailey and Halls (1984) remagnetization circle analysis.

[f]A baked contact test was carried out at this site. B and U refer to baked and unbaked Matachewan dyke rocks, respectively.

[g]This site is reported by Fahrig et al. (1986).

reversely magnetized Fort Frances dykes differ significantly in age. Futhermore, both Fort Frances and Marathon ages are distinctly younger than those of the Biscotasing (2167 ± 2 Ma) and Senneterre (2216^{+8}_{-4} Ma) swarms.

Petrography and geochemistry

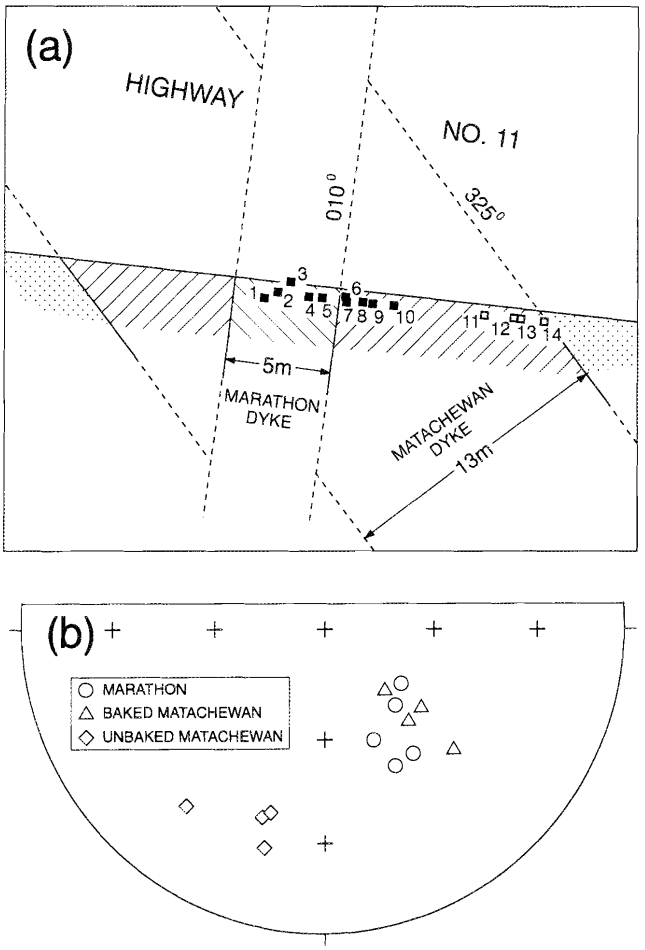
Marathon dykes are quartz tholeiites that consist largely of equigranular to subophitic clinopyroxene and plagioclase in approximately equal proportions. Lesser amounts of amphibole, magnetite, myrmekite, quartz, biotite, and apatite, in approximate order of decreasing abundance, have an aggregate modal content of about 10−15%. The plagioclase is variably altered to saussurite, and the clinopyroxene to turgid aggregates. The pyroxene is locally rimmed by a green amphibole, the development of which increases in proximity to myrmekite and quartz development. Zircon and baddeleyite, with pleochroic haloes, occur in coarse-grained interiors at sites 4, 11, 17−24, 26, and 28, particularly in the amphibole, but also in biotite, indicating that many of the wider Marathon dykes have potential for U−Pb dating. In the coarse-grained interior of one dyke (site 4), rounded, relict olivine occurs within pyroxene. In some sites (2, 12, 13, 17, and 19−21), particularly from the southern end of the swarm, most of the feldspar and about half the pyroxene are altered. In the remaining sites the overall alteration is, on average, about 30%. One site (25) has a conspicuous ophitic texture and is remarkably fresh, with small serpentinized oli-

Table 2. Normal and reverse mean paleomagnetic results for Marathon dykes.

Polarity	No. of sites	D' (°)	I' (°)	k	α_{95} (°)	Plat. (°N)	Plong. (°E)	dm (°)	dp (°)
Normal	16	294.8	57.0	35	6.3	42.7	195.8	9.2	6.7
Reversed	12	133.5	−50.7	42	6.8	51.2	174.7	9.2	6.2

Notes: D' and I', declination and inclination corrected to a common reference site at 49°N, 86°W; k, precision parameter; α_{95}, radius of circle of 95% confidence; Plat. and Plong., coordinates of the paleomagnetic pole; dm and dp, semiaxes of the oval of 95% confidence about the mean paleopole.

Fig. 7. Positive baked contact test at reversely magnetized paleomagnetic site 29. In (*a*), open and closed squares refer to sampling sites at which typical Marathon and Matchewan remanence directions were obtained, respectively. In equal-area plot (*b*), all magnetizations are directed upward.



vine. The Fort Frances swarm to the west (Fahrig 1987; Halls 1986) has similar petrographic characteristics, although comparison of thin sections of similar grain size shows that Fort Frances dykes tend to be greener, reflecting a slightly higher content of amphibole.

Major element geochemistry from chilled-margin samples of Marathon dykes is given in Table 4. Data using an identical sampling strategy and analytical techniques and the same instruments are also available for comparison from the Fort Frances swarm (Southwick and Halls 1987).

In Fig. 10, a Jensen plot (Jensen 1976) is used to compare the Marathon major element geochemical data of this study with the published geochemical data for Fort Frances and Keweenawan dykes. A comparison of the reversely magnetized Marathon dykes and Fort Frances dykes, which are always of reverse polarity, shows that each of these populations generates a linear array on the Jensen plot, reflecting a fractionation trend (cf. Figs. 10*a*, 10*b*). Furthermore, the two arrays are spatially coincident. Both sets of analyses shown in Figs. 10*a* and 10*b* use only fresh, fine-grained samples, less than about 1 m from a dyke margin, and for which the loss on ignition (LOI) is less than about 2% by weight.

The normally magnetized Marathon dykes show a more clustered grouping than the reverse Marathon dykes, with some plotting in the same field as Keweenawan dykes (cf. Figs. 10*a*, 10*c*). The Keweenawan dyke data are more scattered, perhaps reflecting the inclusion of both dyke margin and interior data. However, they have on average more alumina than Marathon and Fort Frances dykes and trend east–west.

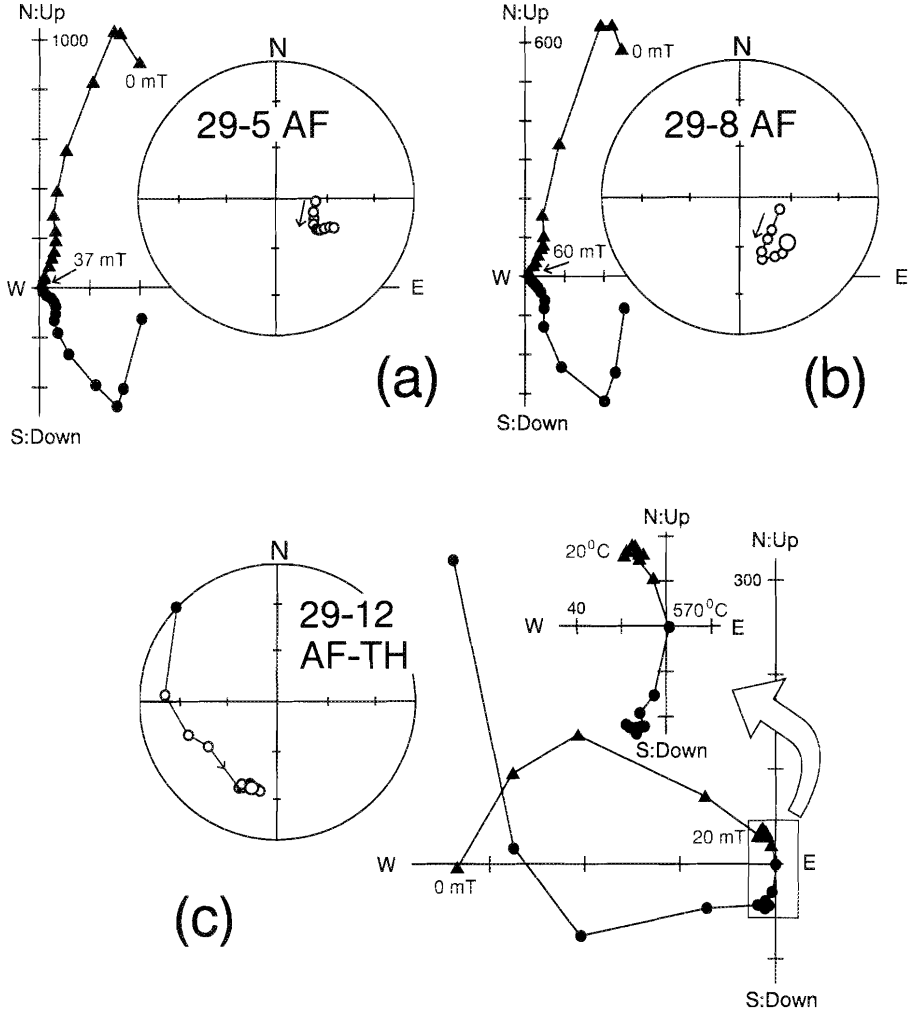
Discussion

The 2121 Ma Marathon and 2076 Ma Fort Frances paleopoles (Fig. 11) continue a northwesterly trend in apparent polar wander for the southern Superior Province, previously established by data for the 2216 Ma Senneterre and 2167 Ma Biscotasing dykes, which were sampled a few hundred kilometres to the southeast (Buchan et al. 1993).

Earlier widely used polar wander paths for North America (e.g., Irving 1979; Hyodo and Dunlop 1989) or the Superior Province (e.g., Irving et al. 1984; Dunsmore and Symons 1990) young from north to south. These earlier paths were poorly constrained, because uncertainties in the age of rock units were typically hundreds of million of years, dating was not carried out at paleomagnetic sampling sites, and paleomagnetic tests were not available to demonstrate primary remanences (Roy 1983; Buchan and Halls 1990). In contrast, with the exception of the normal Marathon pole, the paleopoles shown in Fig. 11 are all primary on the basis of paleomagnetic field tests. In addition, a well-defined U–Pb baddeleyite or zircon–baddeleyite age has been obtained at a paleomagnetic sampling site for each of the units involved, except for the reverse Marathon dykes.

The dramatically different interpretation of the direction of apparent polar wander using the well-constrained data of Fig. 11, compared with that based on the poorly constrained data of earlier polar wander paths, illustrates the importance of collaborative studies of paleomagnetism and U–Pb geo-

Fig. 8. Demagnetization results from baked contact test at site 29. (*a*) Marathon dyke, 70 cm from chilled margin. (*b*) Baked Matachewan dyke, 1 m from chilled margin of Marathon dyke. (*c*) Unbaked Matachewan dyke, 12 m from chilled margin of Marathon dyke. Symbols and explanations as in Fig. 3.



chronology (Buchan and Halls 1990; Buchan et al. 1994). At present, there are no Paleoproterozoic polar wander paths from Precambrian cratons elsewhere in the world that are based on the type of high-precision U–Pb dating of paleopoles described in this study. Until such data become available it will not be possible to compare polar wander paths from different cratons or to establish reliable paleocontinental reconstructions of Paleoproterozoic age based on paleomagnetism.

The stable primary remanence carried by Marathon dykes does not vary systematically in direction along the swarm, suggesting that within the sampling area, no significant crustal tilting related to the 1.1 Ga Mid-Continent Rift has occurred. This inference is consistent with Keweenawan geology. Volcanic rocks exposed on the islands in the southwest corner of Fig. 2 dip less than 10°S, in marked contrast to the eastern margin of the rift where the Superior Province and overlying Keweenawan volcanics have been tilted westwards more than 50° (Halls and Shaw 1988).

As described above, Marathon dykes are divided into two subsets on the basis of magnetic polarity. The normally magnetized subset, dated at 2121 Ma, is 40–50 Ma younger than the 2167 Ma Biscotasing dyke swarm to the east, ruling out the possibility that Biscotasing and normal Marathon dykes form a giant radiating dyke swarm emplaced in a single magmatic event. This conclusion is reinforced by the widely divergent paleopoles obtained from these two set of dykes (Fig. 11). The normal Marathon dykes are also 40–50 Ma older than the 2076 Ma reversely magnetized Fort Frances swarm to the west, ruling out the possibility that these two dyke sets form a giant radiating dyke swarm that was emplaced in a single magmatic event spanning a magnetic field reversal.

The reversely magnetized subset of Marathon dykes has yet to be dated, and hence its relationship to other dykes in the Lake Superior region is uncertain. The small, but statistically significant, asymmetry in the magnetization of normal and reverse Marathon dykes (Fig. 11) may reflect a difference between their emplacement ages. The difference in major element geochemistry of reverse and normal Marathon

Table 3. U–Pb analytical results.

Fraction[a]	Wt. (mg)	U (ppm)	Pb (ppm)[b]	$^{206}Pb/^{204}Pb$ (measured)	Common Pb (pg)[c]	^{208}Pb (%)	$^{206}Pb/^{238}U$[d]	$^{207}Pb/^{235}U$[d]	Correlation coefficient	$^{207}Pb/^{206}Pb$[d]	$^{207}Pb/^{206}Pb$ age (Ma)[e]
Marathon dyke											
A, b, <44, u	0.003	136	52	356	29	1.8	0.3816(0.30)	6.881(0.37)	0.68	0.13077(0.28)	2108.4(9.7)
B, b, <44, u	0.003	317	125	324	65	9.4	0.3690(0.12)	6.552(0.34)	0.64	0.12878(0.27)	2081.4(9.7)
C, b, <44, u	0.003	216	83	444	35	2.3	0.3851(0.16)	6.948(0.22)	0.69	0.13087(0.16)	2109.6(5.6)
D, b, 44–62, u	0.003	1411	533	5 594	18	2.0	0.3807(0.11)	6.859(0.12)	0.94	0.13066(0.04)	2106.9(1.4)
AA, b, 44–72, u	0.012	396	149	5 572	20	2.2	0.3800(0.10)	6.839(0.12)	0.96	0.13054(0.03)	2105.3(1.2)
BB, b, 44–72, u	0.010	323	142	7 079	11	1.9	0.3812(0.10)	6.863(0.11)	0.96	0.13059(0.03)	2106.0(1.1)
Fort Frances dyke											
AA, b, 74–105, u	0.018	172	67	1 942	33	12.0	0.3544(0.09)	6.169(0.10)	0.90	0.12625(0.04)	2046.3(1.6)
B, b, 62–74, u	0.023	208	77	5 222	20	3.4	0.3694(0.23)	6.490(0.22)	0.93	0.12743(0.08)	2062.8(3.0)
E, z, >105, a	0.040	342	169	11 768	26	27.4	0.3683(0.09)	6.473(0.09)	0.98	0.12745(0.02)	2063.1(0.5)
F, z, >105, a	0.041	393	231	21 995	16	39.8	0.3646(0.20)	6.392(0.20)	0.99	0.12717(0.03)	2059.2(1.1)
G, z, >105, a	0.030	504	252	27 424	12	29.5	0.3627(0.08)	6.369(0.08)	0.99	0.12735(0.01)	2061.7(0.4)
H, z, >105, a	0.025	481	249	20 728	13	32.0	0.3632(0.08)	6.354(0.08)	0.99	0.12689(0.01)	2055.3(0.4)

[a]b, baddeleyite; z, zircon; grain size given in μm; u, unabraded; a, abraded.
[b]Radiogenic Pb, corrected for blank, initial common Pb, and spike.
[c]Corrected for spike and fractionation.
[d]Corrected for blank Pb and U and common Pb; errors are in percent at the 1σ level.
[e]Errors are in Ma at the 2σ level.

Fig. 9. (*a*) U–Pb baddeleyite dating at normally magnetized Marathon paleomagnetic site 22. (*b*) U–Pb baddeleyite and zircon dating at a reversely magnetized Fort Frances dyke site on Kakagi Lake. All fractions are baddeleyite, except zircon fractions (*).

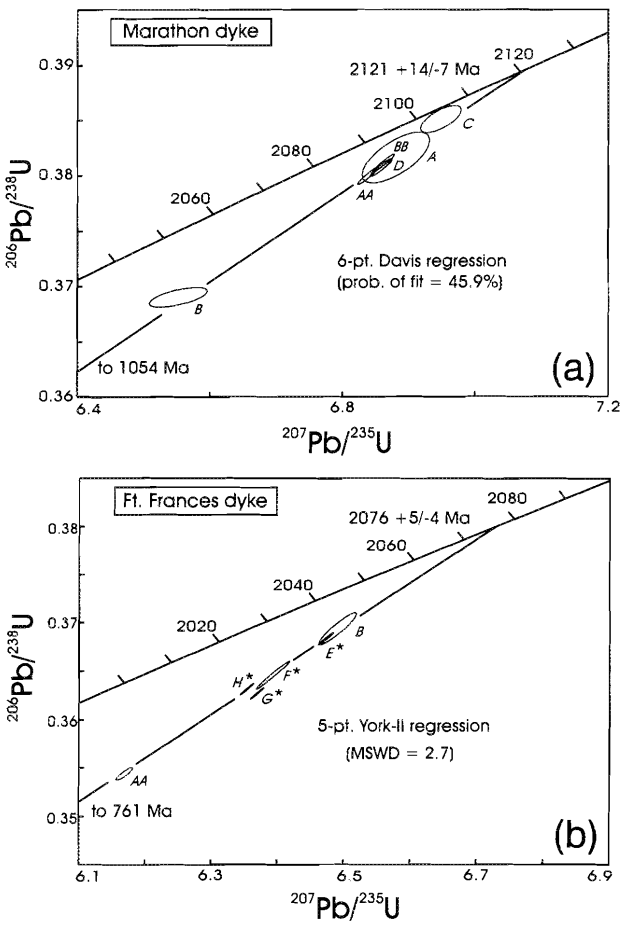


Fig. 10. Jensen diagrams (Jensen 1976) comparing the major element geochemistry of (*a*) chilled margins from the Marathon dykes (each data point represents the average of sample determinations from a site), (*b*) chilled margins from the Fort Frances dykes (from Southwick and Halls 1987), and (*c*) chilled margins and interiors from Keweenawan dykes (based on compilation of Green et al. 1987). Closed and open circles refer to normally and reversely magnetized dykes, respectively. HFB and HMB, high-Fe and high-Mg basalt, respectively.

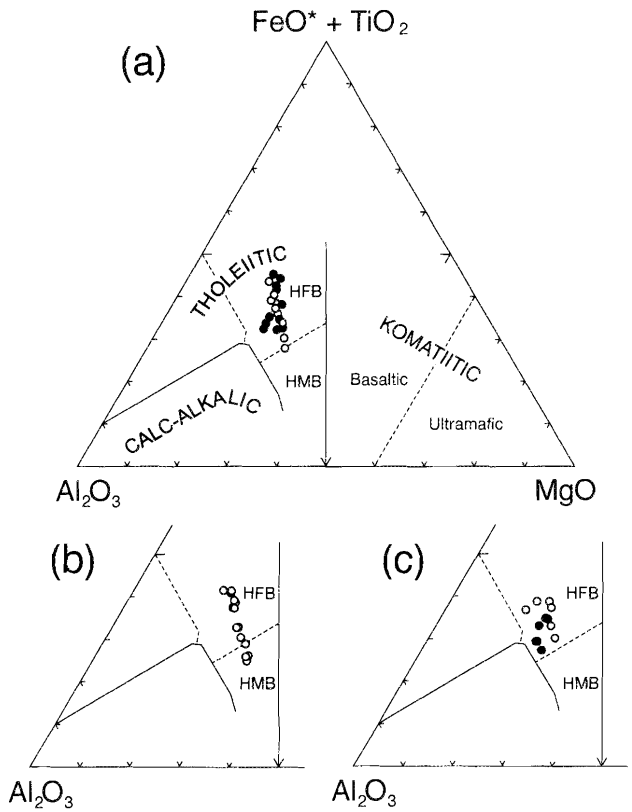


Table 4. Major element geochemistry of Marathon diabase dykes.

Sample[a]	SiO_2	Al_2O_3	CaO	MgO	Na_2O	K_2O	Fe_2O_3	MnO	TiO_2	P_2O_5	LOI	Sum
2-1	49.6	12.7	9.18	5.43	2.46	0.64	15.4	0.24	1.72	0.24	1.54	99.2
2-2	49.8	12.6	8.46	5.35	2.67	0.89	15.1	0.21	1.78	0.25	1.93	98.9
3-1	49.1	12.1	8.10	4.07	2.14	1.33	17.9	0.22	2.78	0.47	0.62	99.0
3-2	49.2	12.1	8.17	6.99	2.23	1.29	18.0	0.22	2.80	0.48	0.16	98.8
5-1	51.3	19.2	9.50	5.14	2.31	0.88	15.5	0.20	1.65	0.19	0.00	99.9
5-2	50.8	13.2	8.83	5.19	2.14	0.84	15.0	0.22	1.63	0.19	1.62	99.7
5-3	50.3	19.1	8.99	5.07	2.36	1.23	15.0	0.17	1.82	0.19	1.93	100.1
6-1	50.5	13.1	10.80	4.80	2.26	0.77	13.6	0.25	1.70	0.20	1.89	99.5
7-1	50.5	13.9	8.40	5.24	2.69	0.87	12.9	0.31	1.73	0.20	2.16	99.0
7-2	50.9	14.1	9.42	4.91	2.16	0.81	14.5	0.22	1.64	0.19	1.31	100.3
8-1	49.0	13.1	8.93	5.30	2.60	0.62	15.4	0.21	2.02	0.24	1.47	99.0
8-2	49.8	13.0	8.64	5.28	2.56	0.94	16.0	0.23	2.01	0.23	1.00	99.8
9-1	49.7	12.8	9.30	4.91	2.46	0.65	16.3	0.29	2.10	0.25	0.31	99.1
9-2	50.1	13.1	9.54	5.13	2.39	0.49	16.2	0.21	2.11	0.25	0.77	100.4
10-1	50.7	12.9	9.61	5.03	2.22	0.62	16.2	0.21	1.79	0.21	0.16	99.7
11-1	49.9	13.0	9.11	4.90	2.98	0.40	15.5	0.19	2.45	0.38	1.08	100.0
11-2	49.5	12.6	9.07	4.70	2.24	0.83	16.9	0.21	2.28	0.35	0.16	98.9
11-3	49.8	12.6	9.04	4.61	2.33	0.69	17.0	0.22	2.34	0.36	1.00	100.1
12-1	48.2	13.6	9.35	5.85	2.44	0.43	15.9	0.22	1.81	0.21	1.39	99.5
12-2	48.2	13.3	9.95	5.82	2.26	0.30	15.4	0.23	1.96	0.22	1.47	99.2
13-1	51.2	13.0	10.20	5.60	2.42	0.28	14.0	0.21	1.37	0.14	0.70	99.2
13-2	51.2	13.1	9.41	5.66	2.73	0.66	13.7	0.21	1.36	0.14	1.23	99.5
13-3	51.1	13.1	9.25	5.76	2.40	0.97	13.8	0.22	1.35	0.14	1.31	99.5
14-1	49.6	13.7	9.04	6.65	2.18	0.77	13.8	0.19	1.21	0.11	1.85	99.2
15-1	50.2	13.0	10.40	5.94	2.29	0.34	14.3	0.22	1.35	0.14	0.77	99.0
15-2	49.8	13.0	10.60	6.44	2.64	0.32	14.1	0.23	1.35	0.15	0.62	99.1
15-3	51.1	13.0	10.70	5.78	2.11	0.29	14.6	0.22	1.38	0.15	0.70	100.1
16-1	49.8	13.4	9.36	5.04	2.34	0.94	15.2	0.21	1.62	0.30	0.70	99.0
16-2	49.9	13.4	9.52	5.09	2.15	0.94	15.3	0.21	1.64	0.30	1.00	99.6
17-1	49.9	12.8	10.30	6.57	1.95	1.89	10.3	0.15	3.21	0.41	1.70	99.3
17-2	47.5	13.3	8.68	4.91	2.15	1.16	16.6	0.24	3.33	0.41	1.39	99.8
19-1	49.8	12.4	9.03	4.70	2.30	0.65	17.2	0.25	2.31	0.35	0.70	99.8
20-1	46.6	11.8	7.94	4.71	2.17	0.99	18.5	0.26	3.66	0.46	1.05	98.2
21-1	47.9	11.9	8.35	4.17	2.33	1.09	18.6	0.25	3.69	0.48	0.75	99.6
23-1	49.0	12.1	8.86	4.70	2.18	0.71	17.4	0.23	2.81	0.33	0.55	98.9
25-1	49.3	14.4	11.20	7.15	2.12	0.23	13.2	0.20	1.01	0.11	0.35	99.3
26-1	48.6	12.0	8.77	4.58	2.11	0.56	17.7	0.24	2.93	0.33	1.15	99.0
28-1	48.9	12.9	9.40	5.24	2.15	0.71	16.2	0.23	1.93	0.25	0.80	98.8
29-1	49.2	13.2	9.90	6.10	2.12	0.59	15.0	0.23	1.78	0.24	0.85	99.3
29-2	51.2	13.5	10.00	5.89	2.22	0.50	15.0	0.23	1.18	0.13	0.20	100.1

Notes: Values in the table are given as weight percents. LOI, loss on ignition.

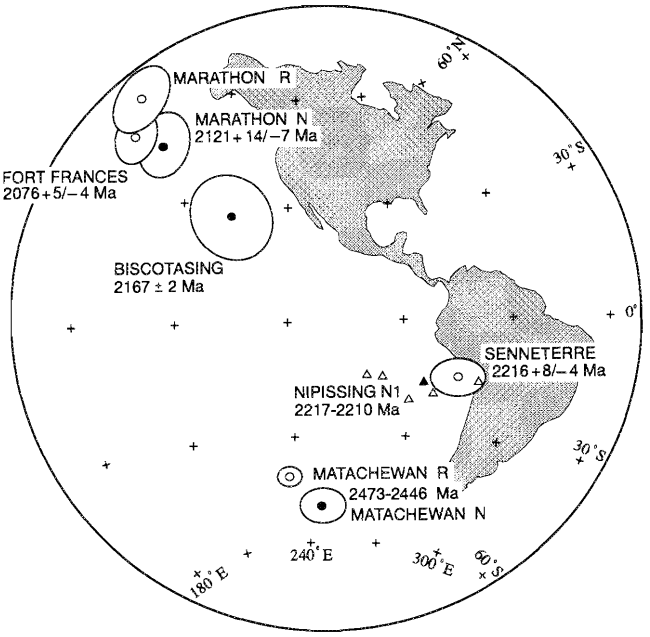
[a]The number before the hyphen refers to the site location as given in Fig. 2, and the number following the hyphen refers to the sample.

dykes also supports a difference in age. The relative age of normal and reverse Marathon dykes is presently unknown. First, they are not observed to cut one another. Second, at the two sites where normal and reversed magnetizations occur together, it is not clear which component is the older.

On the other hand, reverse Marathon and 2076 Ma Fort Frances dykes exhibit significant similarities. The Fort Frances dykes are always of reverse magnetic polarity and yield a paleopole (Halls 1986) intermediate between those obtained for reverse and normal Marathon dykes (Fig. 11). In addition, the similar geochemical signatures of reversely magnetized Marathon dykes and Fort Frances dykes lend support to a genetic link between them. Therefore, our paleomagnetic and geochemical results may indicate that reverse Marathon and Fort Frances dykes belong to a single giant radiating dyke swarm focused south of Lake Superior (Fig. 1). Such radiating swarms, thought to fan from mantle plumes or hot spots, are now recognized in many areas of the world (e.g., Ernst et al. 1996).

Southwick and Day (1983) originally proposed that Fort Frances dykes were related to a hot spot located beneath the future site of the Animikie Basin along the southern margin of the Superior Province. Furthermore, they suggested that the hot spot was the cause of rifting, with the dykes being emplaced along a failed arm of the rift. Our hypothesis that a giant radiating dyke swarm, which consists of both Fort Frances and reversely magnetized Marathon dykes, was associated with a mantle plume centred south of Lake Superior remains to be tested by U–Pb dating of reversely magnetized Marathon dykes.

Fig. 11. Paleomagnetic poles for the southern Superior Province for the 2.47–2.07 Ga period. All poles, with the exception of that for normally magnetized Marathon dykes, have been demonstrated to be primary on the basis of paleomagnetic field tests.



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